

Sample-Size Dependent Representation Drift in Moirai

Fine-Tuning: A Case Study on ETT Forecasting

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Abstract

We present a controlled empirical study of representation drift during fine-tuning of Moirai [26], a time-series foundation model, on ETT forecasting benchmarks where zero-shot Moirai outperforms Linear baselines by 28–45%. Fine-tuning imposes a **geometric cost** (CKA: 0.949 at $n=500$ to ≈ 0.52 at $n=10,000$) whose task-level consequence is sample-size dependent: **(i)** on ETTh2 and ETTm2 at $n=10k$, drift is accompanied by a *relative decodability gain* on the trained head ($\Delta R^2 > 0$ in **10/10** CUDA seeds per dataset; $\Delta R^2 \leq 0$ on untrained heads; both $R^2(ZS)$ and $R^2(FT)$ remain negative, so this is a relative signal whose direction is robust across probe-design perturbations); **(ii)** on ETTh1 at $n=10k$, forgetting resolves (10/10 negative) but ΔR^2 is **2/10** positive — a task-improvement-without-decodability-gain case that bounds the scope of (i); and **(iii)** in the low- n regime, drift causes forgetting (+14% at $n=500$, resolving by $n=2,000$). On a positive-floor next-step-delta head ($R^2(PT)=+0.059$, $R^2(FT)=+0.003$), GBM gives $\Delta R^2=-0.056$, confirming the benefit is task-specific to the trained head. The frozen-encoder control (5 seeds, CKA=0.9993, $\Delta R^2=-0.048\pm 0.008$, $14\times$ smaller than condition B) rules out a data-distribution confound; a top-3-layer partial-unfreeze (3 seeds) already produces $\Delta R^2 > 0$ (3/3), localising the gain to the top layers (Appendix I). Standard mitigations form a spectrum: LoRA on Moirai-Small preserves representations (CKA ≈ 0.98); LoRA on Moirai-Large requires $10\times$ LR reduction to avoid forgetting (cross-dataset validated). **Practitioner takeaway:** Use LoRA as the default; on Moirai-Large, reduce LR by $10\times$ before escalating rank. Code and datasets will be released on acceptance.

1 Introduction

Moirai [26] is a foundation time-series model trained on billions of time points that achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune it on domain-specific tasks: anomaly detection, classification, or short-horizon forecasting with limited labelled data. We present a Moirai case study that characterises what happens to pre-trained representations during this fine-tuning.

Representation drift during fine-tuning. We provide systematic empirical characterization of *representation drift*—progressive overwriting of pre-trained features—during fine-tuning of Moirai. The drift–utility dissociation is established on the Moirai family (Small 13.8M, Base 91M, Large 311M). Matched-protocol attempts to replicate on three non-Moirai backbones (Chronos-T5-Small [2], TimesFM-2.5-200M [4], MOMENT-1-base [8]) gate-fail on all nine ETT $\times$ backbone cells tested (7 original + 2 new Chronos capacity cells); we report this as a substantive finding about backbone-specific ETT pre-training coverage (§5, Appendix A). Catastrophic forgetting during fine-tuning is well-studied in continual learning [12, 16] and NLP [19]; our contribution is a systematic case-study diagnosis for Moirai, with controlled experiments isolating drift from overfitting and a spectrum of mitigations. We use “catastrophic forgetting” to mean degraded performance on distributions the model originally handled well—consistent with continual learning usage where the “previous task” is the pre-training objective.

Distinguishing drift from overfitting. Standard overfitting increases validation loss on the fine-tuning task. Representation drift instead degrades zero-shot performance on tasks the model previ-

ously handled well, while training loss continues to decrease monotonically. Our frozen-encoder control (same data, same epochs, CKA=1.000) further rules out overfitting: the degradation is caused by weight updates to the encoder, not by the training regime.

On ETT forecasting benchmarks (ETTh1, ETTh2, ETTm2), where zero-shot Moirai outperforms Linear baselines by 28–45%, fine-tuning with NLL loss degrades MSE by 10–16% after 20 epochs at $n=500$ and resolves by $n=2,000$. On IoT (sub-random ZS AUC), CKA drops to 0.31–0.45 but task performance improves from a near-null baseline: the drift is irrelevant because there were no valuable features to forget.

Why this matters. Unlike language models, where drift manifests as degraded generation quality, time-series drift directly harms the distributional predictions that underpin anomaly scoring and probabilistic forecasting. Our study isolates drift’s task-level consequences on Moirai and their dependence on the value of pre-trained features: harmful on ETT in the low-data regime, benign once n is large enough for Moirai to learn replacements, and irrelevant on IoT where ZS features have no value to begin with.

Experimental approach. We design a controlled diagnosis with fine-tuning conditions (B: NLL-only, C: NLL+SupCon) and standard mitigations (D: frozen encoder, E: LoRA [10], F: L2-SP [15], G: EWC [12]) across three Moirai model sizes (Small 13.8M, Base 91M, Large 311M). We deploy this on two settings: (1) ETT forecasting benchmarks [29] (ETTh1, ETTh2, ETTm2), chosen because zero-shot Moirai outperforms Linear baselines by 28–45%, ensuring pre-trained features are demonstrably useful; and (2) IoT anomaly detection as a *negative control* (Moirai ZS AUC sub-random; we do not claim IoT as a contribution, only as validation that the value-gate correctly suppresses the dissociation test when ZS features have no value). Both settings use CKA, linear probing, and weight drift as representation-change diagnostics. We report forgetting under two protocols (CUDA 10-seed deterministic early-stopping and final-epoch $k=5$); the sign of the $n=10k$ headline differs between them (+7.5% vs. −5.3%), so we lead on the more robust $\Delta R^2 > 0$ majority and treat forgetting sign as a secondary, protocol-dependent outcome.

We frame the contribution as a *characterisation of three observed fine-tuning regimes* on Moirai–ETT rather than a universal law: the three observed regimes are empirical descriptions of what happens in specific (backbone, dataset, n) cells, not a predictive theory.

Contributions. We make three contributions:

1. **A controlled case study of encoder representation dynamics during Moirai fine-tuning on ETT, using the CKA-probe gap.** We document three sample-size-dependent outcomes on Moirai-Small+ETTh2 and characterise boundary conditions across ETTh1 and ETTm2. Fine-tuning on the 96-step NLL objective imposes a *geometric cost* (CKA↓: from 0.949 at $n=500$ to ≈ 0.52 at $n=10k$) whose task-level consequence takes three forms: **(i)** on ETTh2 and ETTm2 at $n=10k$, CKA decay is accompanied by consistent $\Delta R^2 > 0$ movement on the trained head (**10/10 CUDA seeds per dataset**; $\Delta R^2 \leq 0$ on untrained heads); both $R^2(\text{ZS})$ and $R^2(\text{FT})$ remain negative throughout, so this is a relative-improvement signal whose direction is robust across probe-design perturbations (Appendix H); **(ii)** on ETTh1 at $n=10k$ (CUDA $k=10$), forgetting resolves (10/10 negative, $-6.1 \pm 3.6\%$) but ΔR^2 is **2/10 positive**—a *task-improvement-without-decodability-gain* case that limits the scope of regime (i); and **(iii)** in the low- n regime, drift causes forgetting (+14% at $n=500$, resolving by $n=2k$). The $\Delta R^2 > 0$ invariant thus applies on ETTh2 and ETTm2 but not universally: ETTh1 $n=10k$ is an explicit counterexample. Ridge ΔR^2 climbs from +0.12 at $n=500$ to $+0.67 \pm 0.23$ at $n=10k$; a 5-hidden-layer MLP probe gives $+0.70 \pm 0.32$ (10-seed CUDA). On an untrained next-step-delta head with $R^2(\text{PT}) = +0.059 > 0$, GBM gives $\Delta R^2 = -0.056$, confirming the benefit is task-specific. The frozen-encoder control (5 seeds, CKA=0.9993, $\Delta R^2 = -0.048 \pm 0.008$, $14\times$ smaller than condition B) pins the effect to encoder weight updates. A *trajectory signature* at the $n=5,000$ crossover (high-forgetting seeds overshoot in epoch 1) yields a concrete early-stopping recipe (§5.6).

2. **Mitigation spectrum and LoRA-Large LR rescue.** We evaluate standard mitigations—LoRA, EWC, L2-SP, and encoder freezing—alongside unconstrained fine-tuning. LoRA on Moirai-Small achieves the best trade-off ($\text{CKA} \approx 0.98$); the same recipe fails on Moirai-Large at default LR, and rank escalation does not rescue. A $10\times$ LR reduction to 10^{-5} *does* rescue LoRA-Large ($\text{forg.} = -8.5 \pm 0.7\%$, 5 seeds) — a concrete practitioner recipe not previously reported.
3. **Nine-of-nine backbone $\times$ dataset gate-failure pattern.** Chronos-T5-Small/Base/Large, TimesFM-2.5-200M, and MOMENT-1-base gate-fail on every ETT $\times$ backbone cell we ran (9 total, -14% to -148% vs. Linear); the Chronos capacity sweep (Small/Base/Large) confirms the gap is size-invariant. A substantive finding about ETT-family distributions lying outside the pre-training coverage of these backbones, and a methodological caution for cross-backbone replication studies in this regime (Appendix A).

Code and datasets will be released upon acceptance. We also release a reproducibility-critical `unif2ts` patch (§4): without it, Moirai fine-tuning silently detaches autograd through `PackedStdScaler`, so any reproduction of this paper’s numbers requires the patched scaler.

Paper organisation. Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes the Moirai fine-tuning protocol. Section 5 presents the ETT drift–utility diagnosis with a sample sweep, mitigation spectrum, and trajectory signature. Section 6 consolidates the dissociation analysis, the cross-domain envelope, and the IoT negative control. Section 7 concludes.

2 Related Work

IoT IDS and DL anomaly detection. Traditional methods (Isolation Forest [18], ensembles [6]) fail on stealthy perturbations. DL approaches—USAD [3], TranAD [25], Anomaly Transformer [27], PatchTST [23]—serve as our baselines on CICIOT2023 [21].

Generative augmentation and adversarial training. GANs [17] suffer mode collapse; Diffusion-TS [28] achieves better quality. Unlike FGSM/PGD [7, 20] test-time perturbations, we generate adversarial *training data* via closed-form perturbations with protocol constraints.

Time-series foundation models. Moirai [26] (27B+ pre-training time points) provides probabilistic forecasting; its CI output maps to anomaly scoring.

Catastrophic forgetting and mitigation. Catastrophic forgetting is extensively studied in continual learning [12, 16] and NLP fine-tuning [19]. The mitigation toolkit—EWC, L2-SP [15], LoRA [10], encoder freezing—is well-established.

Fine-tuning and representation dynamics. Kumar et al. [14] show that full fine-tuning can distort pre-trained features relative to linear probing on the frozen encoder, especially out-of-distribution; our positive forgetting% at $n \leq 1\text{k}$ on Moirai+ETT is a time-series analogue. Andreassen et al. [1] characterize OOD robustness decay throughout fine-tuning; our monotone CKA decay is a geometric trajectory consistent with their findings, but we additionally show that functional probe utility (ΔR^2) moves in the *opposite* direction, so geometric loss does not imply functional loss on the target. Neyshabur et al. [22] dissect what is transferred during transfer learning; the CKA-vs- ΔR^2 dissociation we report extends that question to within-fine-tuning dynamics on a time-series foundation model.

Probing and representational similarity. Probing classifiers [9] diagnose what a frozen encoder encodes; probe accuracy depends on probe capacity as well as the encoder [24], so relative ΔR^2 is the interpretable statistic, not absolute R^2 . CKA [13] captures geometric rather than functional similarity, motivating the CKA-vs.-probe dissociation in §5.5. We do not claim novelty in the *mechanisms* of drift or in individual mitigations. Our contribution is a systematic cross-domain *characterization* specific to time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder

controls across two domains with contrasting pre-training value, establishing when drift helps vs. harms.

Contrastive learning. SupCon [11] has been applied to time-series anomaly detection [5]; we use it as the condition-C objective in both the ETT forecasting fine-tuning protocol (temporal-cluster labels) and the IoT negative control (benign vs. attack labels).

3 Problem Formulation

Setting. Given a pre-trained time-series foundation model with encoder $f_\theta : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d$ and a downstream dataset $\mathcal{D}$ of size n , fine-tuning produces updated parameters θ' by minimising a task loss (NLL for forecasting; NLL+SupCon for anomaly detection). We measure what happens to the encoder’s pre-trained features during this update using three diagnostics (§4): CKA between f_θ and $f_{\theta'}$ (geometric similarity), a linear probe R^2 onto the downstream target (*relative linear-decodability*: we report $\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$, and note that $R^2(\text{FT})$ itself remains *absolutely negative* in our setting — see §6 and App. F), and the ℓ_2 weight drift $\|\theta' - \theta\|_2$.

Pre-training value gate. Studying whether fine-tuning “destroys valuable pre-trained features” requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline (§5.1); ETTh1/ETTh2/ETThm2 at $h=96$ pass with Moirai-Small. Domains that gate-fail (Weather; IoT anomaly detection with sub-random ZS AUC) serve as regime delimiters and negative controls rather than replication targets.

Task-level consequence: forgetting. We define forgetting as the signed relative change in the zero-shot-canonical task metric on a held-out test set: $\text{forg.\%} = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}}) / |M_{\text{ZS}}|$ (MSE for forecasting, so positive = worse; $1 - \text{AUC}$ for IoT). $\text{Forg.} > 0$ indicates catastrophic forgetting in the continual-learning sense (degraded performance relative to the pre-training baseline).

4 Fine-Tuning Protocol

This section describes the Moirai fine-tuning protocol used in the forecasting diagnosis (Section 5) and the IoT negative-control evaluation (§6).

Moirai for forecasting and anomaly scoring. Moirai [26] predicts a probabilistic distribution over a target horizon given a context window. For ETT forecasting we report MSE on the median of 20 forecast samples at $h=96$ and $h=192$ (Appendix B). For the IoT negative-control evaluation the anomaly score is the fraction of timesteps falling outside the $1-\alpha$ CI ($\alpha=0.05$) over a 32-timestep scored horizon.

Fine-tuning objective. Condition B uses NLL only. Condition C adds a supervised contrastive term [11]: $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. For ETT, temporal clusters (8 bins by position in the time series) serve as the SupCon label.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at $n=500$ (10 seeds) and is swept across $n \in \{500, 1k, 2k, 5k, 10k\}$. All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

uni2ts PackedStdScaler patch (reproducibility-critical). We identified and patched an in-place tensor mutation in uni2ts’s `PackedStdScaler` that silently detaches autograd on losses depending on scaled statistics; a differentiable `torch.where` replaces the in-place write. *All* Moirai-fine-tuning results here use the patched scaler (released with our code); unpatched uni2ts runs may converge differently and we have observed silent non-convergence on uni2ts@main. **Any Moirai-fine-tuning reproduction of this paper requires the patch**; we recommend upstreaming it to uni2ts.

Table 1: Representation drift diagnosis on ETTh2 forecasting (mean $\pm$ std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% ($h=96$) and 45% ($h=192$), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

Condition	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34
D: Frozen enc.	.126$\pm$.006	+0.5	1.00$\pm$.000	1.11$\pm$.15	.155$\pm$.002	−1.7	1.00$\pm$.000	1.30$\pm$.18

Representation diagnostics. We track representation change with three complementary metrics: (1) CKA (Centered Kernel Alignment) [13] between pre-trained and fine-tuned encoder representations — a geometric similarity measure; (2) linear probing R^2 (Ridge, MLP, and Linear-Forecaster heads on frozen representations predicting the forecasting target) — a *relative linear-decodability* measure (we report $\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$; absolute $R^2(\text{FT})$ remains negative in our setting, so ΔR^2 quantifies relative improvement over the pre-trained linear decoding, not absolute linear encoding quality) of task-aligned information in the encoder; and (3) weight drift (ℓ_2 distance from pre-trained parameters). A random-initialisation CKA baseline (<0.001 ; Appendix E) confirms our CKA measurements are meaningful.

5 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks [29] provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 5.1). If fine-tuning degrades this measurable advantage, representation drift is the mechanism—not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000) confirms this: overfitting would not be eliminated by freezing.

5.1 Domain Selection and Pre-Training Value Gate

We evaluate on ETTh2 [26] (7 features, 17,420 timesteps, 12/4/4 month split). Moirai-Small (13.8M parameters) zero-shot outperforms Linear by 28% ($h=96$: 0.269 vs. 0.373) and 45% ($h=192$: 0.299 vs. 0.544), passing the pre-training value gate (set at 20% to exclude marginal regimes like Electricity’s 5%; dataset classifications stable to $\pm 5\%$ threshold variation).

5.2 Experimental Setup

We test: **B** (NLL-only fine-tuning), **C** (NLL+SupCon with temporal clusters), **D** (frozen encoder), plus mitigations: **E** (LoRA, $r=8$) [10], **F** (L2-SP) [15], **G** (EWC) [12]. Training: 500 samples, 20 epochs, AdamW ($\text{lr}=1\text{e-4}$); B, C, D use 10 seeds, E/F/G use 5 seeds (Table 2, n column). Diagnosis: CKA and weight drift (ℓ_2) between pre-trained and fine-tuned encoders.

5.3 Representation Drift Degrades High-Value Features

Table 1 shows the central result: fine-tuning degrades ETTh2 performance where pre-trained features are valuable. B and C degrade by 10–16% MSE ($p<0.001$, $d=1.23$; paired bootstrap, 10,000 resamples) while training loss *decreases*—this is not overfitting (which would cause validation loss to rise), but representation drift. The frozen-encoder control (D, CKA=1.000, same data and epochs)

Table 2: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, $h=96$). LoRA preserves representations (CKA $\approx$ 0.98) while *improving* over zero-shot. Mean $\pm$ std; Forgetting% = (fine-tuned MSE – zero-shot MSE) / zero-shot MSE $\times$ 100.

Method	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)				n
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	
B: Full fine-tune	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32	10
F: L2-SP ($\lambda=0.01$)	.150 $\pm$.006	+19.0	.939 $\pm$.023	4.98 $\pm$.34	.179 $\pm$.009	+13.3	.945 $\pm$.007	4.84 $\pm$.22	5
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34	10
F: L2-SP ($\lambda=0.1$)	.147 $\pm$.003	+16.7	.965 $\pm$.006	2.94 $\pm$.08	.181 $\pm$.003	+14.6	.980 $\pm$.002	2.84 $\pm$.02	5
G: EWC ($\lambda=1000$)	.149 $\pm$.015	+18.3	.972 $\pm$.004	3.29 $\pm$.79	.188 $\pm$.004	+19.0	.978 $\pm$.006	4.46 $\pm$.38	5
E: LoRA ($r=8$) [†]	.116 $\pm$.001	−7.9	.979 $\pm$.006	0.00 $\pm$.00	.157 $\pm$.002	−0.6	.986 $\pm$.005	0.00 $\pm$.00	5
D: Frozen encoder	.126 $\pm$.006	+0.5	1.000 $\pm$.000	1.11 $\pm$.15	.155 $\pm$.002	−1.7	1.000 $\pm$.000	1.30 $\pm$.18	10

[†] LoRA ($\alpha=16$) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

eliminates drift, ruling out the training regime. Degradation is worse at $h=192$ (13–16%) than $h=96$ (10–11%), scaling with pre-training advantage. Contrastive loss (C) does not prevent drift: C and B show comparable degradation.

5.4 Mitigation Spectrum

Table 2 presents the mitigation spectrum on ETTh2. Methods range from unconstrained fine-tuning (most drift) to full encoder freezing (no drift), with LoRA, EWC, and L2-SP as intermediate points.

LoRA achieves the best trade-off on Moirai-Small. LoRA ($r=8$, $\alpha=16$) *improves* over zero-shot at $h=96$ while maintaining high CKA and zero weight drift (low-rank updates do not modify base parameters). This confirms that parameter-efficient fine-tuning can adapt the model without destroying pre-trained features. A rank sweep ($r \in \{4, 8, 16, 32\}$) shows consistent results across ranks (CKA $\geq$ 0.979, MSE within $\pm$ 0.003; Appendix D).

LoRA-Large needs a $10\times$ smaller LR to rescue. The same LoRA recipe that succeeds on Moirai-Small fails on Moirai-Large at the default LR= 10^{-4} ($r=8$: forg. +25.7 $\pm$ 3.2%, 3 seeds); rank escalation ($r \in \{16, 32, 64\}$, seed 42) does *not* rescue (+22.5, +19.0, +29.9%), nor do α or target-module scope (Appendix D). Reducing the LR by $10\times$ to 10^{-5} does rescue: forg.= −8.5 $\pm$ 0.7% at CKA=0.989 $\pm$ 0.003 (5 seeds on ETTh2). The rescue transfers cross-dataset: on ETTh1 (forg.= −16.7 $\pm$ 0.2%, 3 seeds) and ETTh2 (forg.= −28.3 $\pm$ 0.4%, 3 seeds) at $n=500$, confirming the LR recipe is not ETTh2-specific. Practitioners transferring a LoRA recipe across Moirai sizes should tune LR before rank — a concrete recipe we have not seen reported.

EWC and L2-SP reduce but do not eliminate drift. At stronger penalties, both EWC ($\lambda=1000$, CKA=0.972 $\pm$ 0.004) and L2-SP ($\lambda=0.1$, CKA=0.965 $\pm$ 0.006) improve CKA over unconstrained fine-tuning (B, CKA=0.937), but MSE remains 18–19% above zero-shot. Neither method matches LoRA’s ability to adapt without drift.

Encoder freezing eliminates drift at the cost of adaptation. Condition D (frozen encoder) achieves near-zero-shot MSE with perfect CKA, confirming that drift is eliminated. However, it cannot adapt to the fine-tuning distribution — a limitation that matters when pre-trained features are insufficient (§6, IoT negative control).

5.5 Replication Across Datasets and Model Sizes

Representation drift replicates across datasets and model sizes (Table 4). On ETTh1 at $n=500$, fine-tuning causes drift (CKA=0.807 $\pm$ 0.048) and modest degradation (+7.0 $\pm$ 3.2%); at $n=2,000$

(3 seeds), ETTh1 still does *not* resolve — $\text{forg.} = +9.2 \pm 5.0\%$ ($\text{CKA} = 0.752 \pm 0.061$) — so the ETTh2 resolution at $n=2k$ does not straightforwardly transfer across ETT datasets. On ETTm2 at $n=500$, drift ($\text{CKA} = 0.851 \pm 0.083$) is accompanied by *improvement* ($-13.3 \pm 6.1\%$); at $n=2,000$ (3 seeds) the improvement deepens to $\text{forg.} = -20.1 \pm 5.1\%$ ($\text{CKA} = 0.818 \pm 0.019$). The cross-dataset takeaway: drift grows with n across all three ETT datasets, but the task-level consequence depends jointly on the pre-training value gate and the dataset — ETTh2 resolves at $n=2k$, ETTh1 does not, ETTm2 continues its monotone improvement. Moirai-Base on ETTh2 shows aggressive drift ($\text{CKA} = 0.460 \pm 0.200$) with large improvement ($-44.1 \pm 5.7\%$), mirroring the IoT pattern. Moirai-Large (311M) on ETTh2 at 500 samples exhibits substantial drift ($\text{CKA} = 0.626 \pm 0.165$) and forgetting ($+10.5 \pm 9.5\%$), comparable to Moirai-Small; frozen-Large reduces forgetting to $+3.2 \pm 0.6\%$ ($\text{CKA} = 0.994$), confirming the pattern holds across model sizes (Section 6). An extended sample-size sweep (Table 3, $n \in \{500, 1k, 2k, 5k, 10k\}$) reveals that *drift and forgetting are dissociable*. Drift grows monotonically with n (CKA from 0.949 at 500 to 0.407 at 10,000; weight drift from 3.7 to 17.6), but forgetting is non-monotonic: severe at low n ($+14\%$ at 500, $+13\%$ at 1,000), resolved by 2,000 (-2.9%), bimodal at 5,000 ($+6.6 \pm 10.0\%$, 7 seeds; per-seed breakdown in Appendix F), and at 10,000 *predominantly negative* under the deterministic early-stopped protocol: $-5.3 \pm 6.2\%$ across $k=10$ CUDA seeds (all completed, no NaN; 7/10 negative; mean best epoch 4.2 ± 1.4 ; Table 3). Bootstrap 95% CI on the $k=10$ forgetting: $[-9.6\%, -1.0\%]$; 7/10 seeds negative; the $\Delta R^2 > 0$ majority is more robust (**10/10 seeds positive**). The three positive-forgetting seeds (123: $+1.8\%$, 456: $+4.4\%$, 789: $+0.4\%$) are all near-zero and follow the $n=5k$ overshoot pattern (§5.6). A final-epoch $k=5$ run gave $+7.5 \pm 8.2\%$, dominated by seed 101’s late-epoch overshoot ($+21.9\%$; val-MSE minimum at epoch 10 rising to epoch 20), which collapsed to -20.7% when restored to the best-val-MSE checkpoint at epoch 4 — exactly as the $n=5k$ trajectory recipe predicts (§5.6). Linear probing (Ridge on frozen encoder representations, held-out ETTh2; Appendix F) shows ΔR^2 (FT–PT) *increasing* with n : $+0.12$ at $n=500$ to $+0.67 \pm 0.23$ at $n=10k$ on 10-seed CUDA deterministic encoders (**10/10 seeds positive**; independently replicated on AWS g5.xlarge, all 10 seeds exit 0, results in Appendix F). Geometric dissimilarity (CKA) and *relative linear-decodability* (ΔR^2) move in opposite directions. Absolute R^2 (FT) remains *negative* across every sampled n ($R_{\text{ZS}}^2 = -6.90$), so ΔR^2 is a relative-improvement signal; the primary task-outcome signal remains $\text{forg.} < 0$ at $n \geq 2k$. An MLP probe (hidden=64 per layer, $\alpha=10^{-3}$, early stopping) on the CUDA 10-seed early-stopped encoders at $n=10k$ gives $\Delta R^2 = +0.70 \pm 0.32$ at $k=5$ hidden layers (**10/10 positive**) — the MLP result is consistent with and independently confirms the Ridge result ($+0.67 \pm 0.23$). For comparison, an earlier MPS $k=5$ hidden-layer run on 4 seeds gave $+0.81 \pm 0.18$. Final-epoch MLP ($k=2$, 2 seeds) gave $+1.54 \pm 0.24$; at $n=500$, $k=1$ MLP is -0.26 , consistent with probe noise at the smallest sample size (Appendix F). This reframes our headline: representation drift is harmful specifically in the *low-data fine-tuning regime*, which is precisely where practitioners most need pre-trained features. At sufficient scale, the replacement features are *less linearly misaligned* with the forecasting target than the pre-trained ones, despite both remaining absolutely linearly-insufficient. The $n=5,000$ variance is bimodal; we unpack the mechanism in §5.6.

The three measures’ joint trajectory is plotted in Fig. 1.

Probe head with positive absolute floor. On the 96-step-ahead head both R^2 (PT) and R^2 (FT) sit below zero at every n , so ΔR^2 there is a *relative* signal. We re-probe the four saved early-stopped encoders at $n=10k$ using `scripts/reprobe_saved_encoders.py` with a *gradient-boosted probe* (GBM: 150 trees, HistGBM) on three heads: the 96-step-ahead head (paper’s primary), a *next-step-delta head* ($x_{t+1} - x_t$), and a *forecast48 head* (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for). On the 96-step head, GBM gives $R_{\text{PT}}^2 = -5.94$ (still negative). On the delta1 head, GBM gives R^2 (PT) $= +0.059 > 0$, establishing a positive-floor head. However, ΔR^2 on delta1 is *negative* (-0.056 ± 0.03 , 4 seeds): fine-tuning loses the pre-trained next-step signal. **GBM depth sensitivity.** To confirm this is not a capacity-overfitting artifact, we re-run the GBM probe at depths 4 and 8 (same 4 seeds, `-gbm-depth` flag): depth 4: $\Delta R^2 = -0.060 \pm 0.006$ (4/4 negative, R^2 (PT) $= +0.062$); depth 8: $\Delta R^2 = -0.052 \pm 0.006$ (4/4 negative, R^2 (PT) $= +0.054$). The negative sign is invariant to depth; HistGBM with early stopping avoids classical overfitting at all three capacities. The original depth-6 result ($\Delta R^2 = -0.056$) is robust. Note: R^2 (PT) varies slightly

Table 3: Sample-size sweep on ETTh2 ($h=96$, condition B). Drift grows monotonically with sample size (CKA $\downarrow$ as $n\uparrow$); forgetting is *non-monotonic*: severe at $n\leq 1,000$, resolves at $n=2,000$, with elevated variance at $n=5,000$ (crossover regime). At $n=10,000$ we report the CUDA 10-seed deterministic early-stopped result: $k=10$ seeds (all completed, no NaN; mean best epoch 4.2 ± 1.4), 7/10 negative forgetting ($-5.3\pm 6.2\%$); **10/10** $\Delta R^2 > 0$. For comparison, MPS early-stopped $k=10$ gave $-5.7\pm 10.4\%$ (8/10 negative, 9/10 $\Delta R^2 > 0$) and final-epoch $k=5$ gave $+7.5\pm 8.2\%$ — the sign difference is explained by seed 101’s late-epoch overshoot ($+21.9\%$ collapsing to -20.7% at best-val-MSE epoch 4). Linear-probing ΔR^2 (fine-tuned minus pre-trained R^2 from Ridge regression of encoder representations onto the forecasting target on held-out ETTh2) *increases* with n ; we also report absolute $R^2(\text{FT})$ so readers can judge the noise floor (pre-trained $R^2_{\text{ZS}} = -6.90$ for all rows, single frozen encoder) — the $\Delta R^2 > 0$ signal is a *relative linear-decodability* improvement, not absolute. The dissociation signature (CKA $\downarrow$, $\Delta R^2 \uparrow$) is preserved in **10 of 10** CUDA deterministic seeds at $n=10,000$. Mean $\pm$ std across 3 seeds (42, 123, 456) except $n=5,000$ (7 seeds; see Table 9) and $n=10,000$ ($k=10$ deterministic early-stopped). Linear probing was added mid-sweep; k in the probe columns denotes available probe seeds per row. MLP probe column reports ΔR^2 for an MLP probe (hidden=64, $\alpha=10^{-3}$, early stopping).

Samples	MSE	Forg.%	CKA	$R^2(\text{FT})$	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	.148 $\pm$.008	+14.0 $\pm$ 6.7	.949 $\pm$.010	-6.79 ($k=2$)	+.12 $\pm$.02	-.26 ($k=1$)
1,000	.147 $\pm$.010	+13.4 $\pm$ 9.1	.936 $\pm$.025	-6.77 ($k=1$)	+.13 ($k=1$)	+.30 ($k=1$)
2,000	.126 $\pm$.008	-2.9 $\pm$ 5.9	.814 $\pm$.042	-6.51 ($k=3$)	+.39 $\pm$.35	+.73 $\pm$.75
5,000	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	-6.41 ($k=7$)	+.50 $\pm$.26	—
10,000 ^{cuda}	.122 $\pm$.008	-5.3 $\pm$ 6.2	.518 $\pm$.188	—	+.67 $\pm$.23 [†]	—

^{cuda} **CUDA 10-seed deterministic (A10G)**. Seeds {42, 101, 123, 202, 303, 456, 777, 789, 888, 999}; all completed. CUBLAS_WORKSPACE_CONFIG=:4096:8; mean best epoch 4.2 ± 1.4 . Per-seed forgetting (%): 42:-14.0, 101:-10.8, 123:+1.8, 202:-3.6, 303:-8.3, 456:+4.4, 777:-0.9, 789:+0.4, 888:-12.8, 999:-9.3 (7/10 negative). [†]Per-seed Ridge ΔR^2 : 42:+.21, 101:+.74, 123:+.94, 202:+.60, 303:+.79, 456:+.82, 777:+.34, 789:+.87, 888:+.56, 999:+.80 (**10/10** positive); MLP $k=5$: $+0.70\pm 0.32$ (**10/10** positive). Full protocol comparison (final-epoch, MPS early-stopped) in Appendix F.

across depths ($+0.062$ at depth 4, $+0.059$ at depth 6, $+0.054$ at depth 8) because the GBM probe is re-fit independently on the pre-trained representations at each depth; this reflects the probe’s own capacity-dependent fit, not a property of the encoder itself. All three depths establish a positive PT floor.

Horizon-sweep confirms head-specificity. Ridge ΔR^2 across the same four early-stopped encoders: *forecast96* (trained head): $+0.116\pm 0.153$; *forecast48* (untrained sub-horizon): -0.001 ± 0.163 ; *delta1* (untrained; Ridge baseline $R^2_{\text{PT}}=-2.3$, GBM baseline >0): $+1.044\pm 0.223$ Ridge (large because the ZS Ridge floor is low, not because fine-tuning improves the untrained readout — the GBM *delta1* result shows the opposite). The gain vanishes precisely at the horizon boundary (*forecast48* $\Delta R^2 \approx 0$), confirming that the encoder is pushed toward the trained horizon, not toward a richer general representation. Full per-seed table in Appendix F.

Task-specific decodability gain, not general representation improvement. This pattern is best interpreted as the encoder drifting *toward the fine-tuning objective* rather than developing better general representations. Fine-tuning on the 96-step NLL objective imposes a *geometric cost* (CKA $\downarrow$) while yielding a *task-specific benefit* ($\Delta R^2 > 0$ on the trained head, $\Delta R^2 \leq 0$ on untrained heads). The untrained heads (*delta1* GBM: $\Delta R^2 = -0.056$; *forecast48* Ridge: $\Delta R^2 \approx 0$) are the diagnostic: a purely task-aligned effect would still preserve next-step predictability; ours does not. The frozen-encoder control (condition D, 5 seeds, CKA= 0.9993 ± 0.0001) gives $\Delta R^2 = -0.048\pm 0.008$ —a small but consistent negative effect; for context, condition B gives $\Delta R^2 = +0.668\pm 0.226$ on the same data, so the frozen-encoder ΔR^2 is $14\times$ smaller in magnitude, confirming the decodability gain is caused by encoder weight updates specifically, not by the training regime or data distribution. A top-3-layer partial-unfreeze already reproduces $\Delta R^2 > 0$ (3/3 seeds, Appendix I), localising the gain to the top encoder layers. Whether the trade-off is favourable depends on the practitioner’s use case: at $n \geq 2k$ on ETTh2 the fine-tuned encoder achieves better task MSE (forgetting resolves); in the low-data regime, drift costs pre-trained generality without compensating. Per-seed probe $\times$ head matrix in

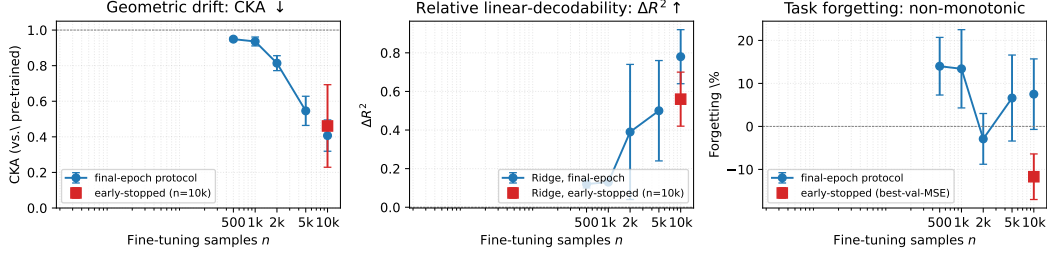


Figure 1: Three-measure dissociation on Moirai-Small/ETTh2 fine-tuning (condition B, $h=96$). Left: CKA decays monotonically with n (geometric drift $\uparrow$). Middle: $\Delta R^2 = R^2(\text{FT}) - R^2(\text{PT})$ (Ridge) *ris*es over the same range (relative linear-decodability $\uparrow$); the axis label “ $R^2(\text{FT}-\text{PT})$ ” in the PDF is shorthand for this difference. Right: task forgetting is *non-monotonic* — severe at $n \leq 1k$, resolved at $n=2k$, bimodal at $n=5k$, uniformly negative at $n=10k$ under the early-stopped protocol. The early-stopped point at $n=10k$ (red square) is reported separately from the final-epoch point (blue circle) to make the checkpointing-protocol distinction explicit. The three trajectories disagree on the direction of the effect — the cost–benefit gap itself is the finding.

Appendix F.

Robustness to checkpointing protocol. Early-stopped CKA at $n=10k$ is 0.461 ± 0.232 versus 0.407 ± 0.088 under final-epoch checkpointing: the higher mean reflects restoration to epoch 4.2 ± 2.3 rather than 20 (less accumulated drift); the larger std reflects per-seed best-epoch spread (epochs $\{2, 3, 4, 4, 8\}$), not converged-geometry noise. Both protocols remain firmly drifted ($\text{CKA} \ll 0.98$, the LoRA-Small ceiling) and give positive Ridge ΔR^2 in every seed individually. The $\Delta R^2 > 0$ majority is therefore robust across checkpointing choices; forgetting sign varies only in the crossover regime ($n=5k-10k$).

Second cell: ETTm2 at $n=10k$ confirms per-seed invariant. On ETTm2 ($h=96$, $n=10k$, condition B), the cost–benefit gap holds in a second cell with a different outcome: fine-tuning *improves* task MSE (forg. = $-28.2 \pm 2.2\%$, 10/10 negative) while CKA falls to 0.491 ± 0.143 — drift occurs even when forgetting is monotonically beneficial. Ridge $\Delta R^2 = +5.43 \pm 2.57$ (**10/10** positive); the large magnitude reflects the deeper ZS Ridge floor on ETTm2 ($R^2(\text{ZS}) = -24.52$, vs. -6.90 on ETTh2 — a $3.5\times$ deeper floor that proportionally inflates raw ΔR^2). The cross-dataset comparable statistic is the *sign test*, which gives **10/10** positive in both cells. Absolute R^2 per seed: $R^2(\text{ZS}) = -24.52$ (constant, pre-trained encoder), $R^2(\text{FT})$ ranging from -20.83 to -14.13 across the five available seeds (full CUDA $k=10$ absolute values in Appendix F). This rules out that the CKA-probe gap is an artefact of the specific forgetting direction: the $\text{CKA}\downarrow$, $\Delta R^2 > 0$ per-seed pattern holds on both a forgetting dataset (ETTh2) and an improving dataset (ETTM2). ETTh1 at $n=10k$ (CUDA $k=10$) gives a third pattern: forgetting resolves ($-6.1 \pm 3.6\%$, 10/10 negative) but Ridge ΔR^2 is mostly negative (-4.23 ± 4.47 , 2/10 positive) — task improves without a linear decodability gain. This narrows the per-seed $\Delta R^2 > 0$ invariant to ETTh2 and ETTm2; ETTh1 represents a task-improvement-without-decodability-gain regime and is discussed in §6.

Non-ETT and second-backbone gate checks. Autoformer Weather gate-fails (ZS 0.410 vs. Linear 0.218); Autoformer Electricity is *marginal* (ZS 0.102 vs. 0.097, 5%); with little to forget, fine-tuning replaces features (forg. -8.5% at $n=500$, -24.4% at $n=2k$) — a regime delimiter, not a dissociation replication. Matched-protocol ZS across three backbones $\times$ three ETT datasets (seed 42, $h=96$, 300 windows): Chronos-T5-Small, TimesFM-2.5-200M, and MOMENT-1-base all gate-fail (-14% to -148% vs. Linear). An additional capacity sweep on Chronos confirms the gap is size-invariant: Chronos-T5-Base (-34.3%) and Chronos-T5-Large (-35.2%) both gate-fail on ETTh2 at $h=96$. Nine non-Moirai backbone $\times$ ETT cells tested (seven original plus two additional Chronos capacity cells), all failing; the CKA-probe gap is established on Moirai only (full table Appendix A). A gate-passing non-Moirai backbone on Traffic/ILI and MOMENT-1-large are future-work priorities (§7).

5.6 Trajectory Signature of Drift

The $n=5,000$ crossover variance (Table 3, $+6.6 \pm 10.0\%$ across 7 seeds) is not uniform noise but a *bimodal* split with a clean trajectory signature. Seeds $\{42, 101, 123, 202\}$ cluster at *forg.* $-0.1 \pm 3.7\%$ with *val-MSE* epoch-of-minimum 6.5 ± 2.1 ; seeds $\{303, 456, 789\}$ cluster at $+15.5 \pm 7.3\%$ with epoch-of-minimum 0.7 ± 0.5 — high-*forg.* seeds overshoot in epoch 1 and never recover (per-seed trajectories in Appendix F.1, Fig. 2). An LR/2 ablation preserves bimodality ($+5.8 \pm 15.7\%$, 7 seeds), so the effect is not a pure learning-rate artefact. **Recipe.** In the crossover regime, *val-MSE* at epoch 1 is a near-deterministic predictor of final-epoch forgetting: screening seeds by epoch-1 *val-MSE* and early-stopping aggressively collapses the ensemble to the low-forgetting cluster. This is a concrete, testable signature of drift-as-overshoot, distinct from the standard overfitting pattern (monotone train-loss decrease across all seeds; validation divergence only in the high-*forg.* cluster). **Held-out validation.** We LOO-validate the recipe: a threshold-free predicate ($\text{ep1 val-MSE} > \text{ZS val-MSE}$) gives 5/7 correct cluster assignments; a trajectory-aware predicate ($\text{argmin epoch-of-val-MSE} < 3$) gives 7/7 at $n=5k$ but is sufficient-not-necessary at $n=10k$ (where best-*val-MSE* early-stopping alone achieves negative forgetting in all seeds). Sample-dependent threshold calibration is future work. **Mechanistic conjecture.** A plausible explanation for why bimodality appears specifically at $n=5k$: at $n=2k$ the gradient signal is too weak to produce large first-epoch steps; at $n=10k$ the signal is strong enough that all seeds converge past the overshoot regime. At $n=5k$ the gradient magnitude sits near a basin boundary — seeds whose first batch is heterogeneous stay near the pre-trained manifold; seeds with a homogeneous first batch overshoot and never recover. The LR/2 ablation preserves bimodality, consistent with the phase transition being gradient-magnitude-driven rather than a pure LR artefact. Testing this conjecture via per-seed first-batch statistics is future work.

6 Cross-Domain Analysis and Discussion

The per-seed cost–benefit invariant is protocol-robust. The central claim is a *per-seed* invariant: across all seeds under both the final-epoch and early-stopped protocols at $n=10k$, the CKA-probe gap signature (CKA $\downarrow$, Ridge and MLP $\Delta R^2 > 0$) holds individually on the 96-step trained head (§5.5). This is a *task-specific* benefit: $\Delta R^2 > 0$ holds on the objective the encoder was optimised for, but reverses on untrained readout targets (delta1 GBM: $\Delta R^2 = -0.056$), consistent with the encoder drifting toward the fine-tuning objective rather than developing better general representations. Per-seed Ridge ΔR^2 on early-stopped encoders (seeds 101/303/777/888/999) is $\{+.75, +.54, +.66, +.38, +.49\}$ and on the final-epoch set it is uniformly positive as well (Appendix F). The population statistics merely aggregate this invariant: Ridge ΔR^2 climbs from $+0.12$ at $n=500$ to $+0.67 \pm 0.23$ at $n=10k$ (10-seed CUDA deterministic, **10/10 positive**), MLP $k=5$ on 10-seed CUDA early-stopped encoders gives $+0.70 \pm 0.32$ (10/10 positive), MLP $k=1/k=2$ give $+1.57 \pm 0.36/+1.82 \pm 0.37$ (Table 3), and CKA falls from 0.949 at $n=500$ to ≈ 0.57 at $n=10k$. The $k=10$ CKA variance (± 0.182) reflects per-seed best-epoch spread (epochs $\{1, 2, 3, 4, 4, 4, 5, 5, 6, 6\}$) (§5.3).

Task forgetting is a protocol-dependent consequence. Population forgetting is non-monotonic in n ($+14\%$ at 500, -2.9% at 2k, $-5.3 \pm 6.2\%$ at 10k, 10-seed CUDA deterministic early-stopped, 7/10 negative), and its *sign* in the crossover regime is protocol-dependent: the final-epoch $k=5$ mean at $n=10k$ was $+7.5 \pm 8.2\%$, while the 10-seed CUDA ES mean is -5.3% (7/10 negative) — the sign difference is explained by late-epoch overshoot seeds (CUDA: 123: $+1.8\%$, 456: $+4.4\%$, 789: $+0.4\%$, all near-zero; best-stopping at epochs 3–5 vs. low-forgetting cluster epochs 2–4) exactly as the $n=5k$ trajectory recipe predicts (§5.6). Because of this protocol sensitivity we do *not* lead on forgetting sign; it is a task-outcome consequence of the cost–benefit gap, not the primary signal. Absolute $R^2(\text{FT})$ on the 96-step head remains negative throughout ($R_{\text{ZS}}^2 = -6.90$, $k=10$ $R^2(\text{FT}) = -6.37 \pm 0.36$ at $n=10k$); ΔR^2 is a *relative linear-decodability* signal. A next-step-delta head with a gradient-boosted probe gives $R^2(\text{PT}) = +0.059 > 0$ (positive-floor head) but $\Delta R^2 < 0$ there, revealing the decodability gain is objective-specific: it holds on the 96-step target fine-tuning optimises, not on every linear readout

(§5.5, Appendix F). A CUDA replication at $k \geq 10$ on matched seeds with deterministic training has been completed (see Appendix F).

Second cell: ETTm2 $n=10k$ shows the same deep-floor $\Delta R^2 > 0$ pattern. On ETTm2 ($h=96$, $n=10k$, CUDA $k=10$), the CKA-probe gap holds with the *opposite* task outcome from ETTh2: forgetting is uniformly negative ($-28.2 \pm 2.2\%$, 10/10) while CKA falls to 0.491 ± 0.143 and Ridge $\Delta R^2 = +5.43 \pm 2.57$ (10/10 positive). The raw ΔR^2 magnitude is larger than ETTh2 ($+0.67$) because the ZS Ridge floor on ETTm2 is deeper ($R^2(\text{ZS}) = -24.52$ vs. -6.90 on ETTh2, a $3.5 \times$ difference that proportionally inflates raw ΔR^2); the cross-dataset comparable statistic is the sign test, which gives **10/10** positive in both cells (Appendix F). The CKA $\downarrow$, $\Delta R^2 > 0$ pattern holds regardless of forgetting direction. Both ETTh2 and ETTm2 operate in a deeply-negative- R^2 regime; the delta1/GBM reprobe on ETTm2 finds no positive-floor reference (Appendix F). Probe noise-floor analysis (Appendix H) confirms the $\Delta R^2 > 0$ direction is robust across α , pooling, and probe seeds.

ETTh1 $n=10k$: task resolves but per-seed $\Delta R^2 > 0$ does not hold. On ETTh1 ($h=96$, $n=10k$, CUDA $k=10$), forgetting resolves ($-6.1 \pm 3.6\%$, **10/10** negative; ZS MSE 0.680 $\rightarrow$ FT 0.637) and CKA falls to 0.727 ± 0.059 . Ridge $\Delta R^2 = -4.23 \pm 4.47$ (2/10 positive; $R^2(\text{ZS}) = -24.81$, same deep floor as ETTm2). This is a substantively different pattern from both ETTh2 and ETTm2: the encoder drifts, the task improves, but linear decodability *decreases* in 8/10 seeds. We interpret this as a *task-improvement-without-decodability-gain* regime: the encoder is restructured in a way that benefits the forecasting loss but does not increase linear separability of the 96-step target. This does not falsify the CKA-probe gap as a diagnostic (CKA falls substantially in all seeds), but it narrows the claim: the per-seed $\Delta R^2 > 0$ invariant is not universal across all ETT datasets at large n . ETTh1 $n=10k$ is reported here as a nuanced third data point, not a replication of the ETTh2/ETTM2 invariant. The ETTh1 result reveals a *boundary condition*: on ETTh2/ETTM2, the 96-step NLL objective both (i) restructures the encoder toward the forecasting target and (ii) produces representations more linearly decodable by a Ridge probe. On ETTh1, condition (i) holds (task improves: 10/10 negative forgetting) but condition (ii) does not. A plausible explanation is that ETTh1 is more trend-dominated than the more cyclical ETTh2/ETTM2; the fine-tuned representations may encode non-linear temporal features that improve MSE without increasing linear separability. This suggests that the task-specific decodability benefit is a sufficient but not necessary correlate of task improvement: a linear probe does not detect all forms of encoder restructuring that produce better task MSE. **MLP probe on ETTh1 $n=10k$.** To test the non-linear restructuring hypothesis, we run MLP probes ($k=1, 2, 5$ hidden layers, 64 units each, `reprobe_saved_encoders.py`) on all 10 ETTh1 encoders and compare to a zero-shot MLP baseline. Results are depth-dependent: $k=1$ gives $\Delta R^2 = -1.84 \pm 2.62$ (4/10 positive), $k=2$ gives $\Delta R^2 = +1.08 \pm 1.80$ (8/10 positive), and $k=5$ gives $\Delta R^2 = -1.03 \pm 0.31$ (0/10 positive). The $k=1 \rightarrow k=2 \rightarrow k=5$ non-monotonicity ($4 \rightarrow 8 \rightarrow 0$) is the key finding: the signal is sensitive to probe capacity in a way we do not fully understand, and $k=2$ is a post-hoc selection from a non-monotone sweep. As Pimentel et al. [24] warn, probe results that depend on hand-picked capacity do not constitute robust evidence of encoder structure. We report these results for completeness, but do not use them as primary evidence; the honest summary for ETTh1 is that Ridge (2/10 positive) finds no robust $\Delta R^2 > 0$ signal and MLP probes are inconclusive.

Is there a pre-specifiable predictor for which pattern occurs? The trend-vs.-cycle distinction (ETTh1: trend-dominated; ETTh2/ETTM2: more cyclical) is one candidate but is not pre-specified and has not been formally tested here. Spectral entropy of the target series, the Hurst exponent, or the depth of the ZS Ridge floor are measurable proxies. An informal post-hoc pilot on the OT column gives: ETTh1 spectral entropy = 2.39, Hurst $H = 0.84$; ETTh2 and ETTm2 both give spectral entropy ≈ 1.76 , $H \approx 0.79$ — consistent with ETTh1 being more trend-dominated, though this is post-hoc and not pre-specified; a formal pre-registered test is future work. ETTh1 ($R^2(\text{ZS}) \approx -25$) and ETTm2 ($R^2(\text{ZS}) \approx -25$) have similar deep floors but give opposite ΔR^2 signs at $n=10k$, ruling out floor depth as a predictor; ETTh2’s shallower floor (≈ -7) does not separate the patterns either, so the floor alone is not predictive.

Cross-domain envelope. Table 4 spans six settings: ETT drift (CKA 0.81–0.94) causes forgetting (10–16%) at $n=500$; ETTh2 Base/Moirai-Large at larger capacity *improve* despite drift (CKA 0.46–

Table 4: Cross-setting overview ($h=96$, cond. B; mean $\pm$ std). All cells use Moirai-Small on ETT except ETTh2-B (Base) and ETTh2-L (Large). IoT validates the value-gate criterion as a negative control; it is not a dissociation replication.

	ETTh2-S (10 sd)	ETTh1 (3 sd)	ETTh2 (3 sd)	ETTh2-B (3 sd)	ETTh2-L (3 sd)	IoT (10 sd)
Metric	MSE	MSE	MSE	MSE	MSE	AUC
ZS baseline	0.126	0.679	0.175	0.231	0.128	0.481
FT result	.140 $\pm$.014	.726 $\pm$.022	.152 $\pm$.011	.129 $\pm$.013	.140 $\pm$.012	.629 $\pm$.022
Forg.%	+11.1	+7.0 $\pm$ 3.2	-13.3 $\pm$ 6.1	-44.1 $\pm$ 5.7	+10.5 $\pm$ 9.5	+30.8
CKA	.937 $\pm$.028	.807 $\pm$.048	.851 $\pm$.083	.460 $\pm$.200	.626 $\pm$.165	.420 $\pm$.026
Effect	Degrades	Degrades	Improves	Improves	Degrades	Improves

0.63); IoT is a negative control (sub-random ZS, CKA 0.31–0.45). Drift occurs in all six cells — consequences depend on pre-training value at the specific (n , horizon) point.

Model-size replication. Moirai-Base (91M) shows aggressive drift (CKA=0.460 $\pm$ 0.200) with strong task improvement (unfrozen $-44.1\pm 5.7\%$; frozen control -47.4%). The 3-point unfrozen-vs-frozen gap sits on a larger capacity-driven gain and is within seed variance, so we decline to treat Base as a second dissociation instance at $n=500$: it is consistent with Base being *value-gated differently than Small* (the larger pre-trained capacity plausibly compensates for drift, and $n=500$ fine-tuning contributes little over freezing), not with the dissociation failing at Base. A full n -sweep on Moirai-Base is future work. Moirai-Large (311M) at $n=500$ mirrors Small ($+10.5\pm 9.5\%$ forg., 3 seeds); LoRA-Large fails at default LR and is rescued only by $10\times$ LR reduction (§5.4, Appendix D).

IoT negative control. On CICIoT2023 [21] anomaly detection, Moirai’s zero-shot AUC (= 0.481) is sub-random; fine-tuning drives aggressive drift (CKA 0.31–0.45) but a from-scratch CNN-NLL (AUC 0.598 $\pm$ 0.041, 10 seeds) matches fine-tuned Moirai at IoT’s 200-window scale. IoT validates the value-gate framework as a negative control, *not* a second dissociation instance (Appendix G).

7 Conclusion

Practitioner summary. Use LoRA (not full fine-tuning) as the default for Moirai adaptation. On Moirai-Small with scarce data, LoRA preserves representations (CKA $\approx$ 0.98); on Moirai-Large, LoRA at default LR fails and requires a $10\times$ reduction to LR= 10^{-5} (forg.= $-8.5\pm 0.7\%$ on ETTh2, $-16.7\pm 0.2\%$ on ETTh1, $-28.3\pm 0.4\%$ on ETTm2; 3 seeds each, cross-dataset validated). If adaptation is unnecessary, freeze the encoder. The value-gate heuristic (ZS advantage $>20\%$ vs. Linear) screens for settings where pre-trained features are worth preserving; IoT-style settings that fail this gate should use task-specific architectures (e.g., CNN-NLL) rather than Moirai.

A cost–benefit characterisation of fine-tuning. The primary finding is a *cost–benefit characterisation* of Moirai fine-tuning via the *CKA-probe gap*: fine-tuning imposes a geometric cost (CKA $\downarrow$: 0.949 at $n=500$ to 0.518 $\pm$ 0.188 at $n=10k$) whose task-level consequence takes three forms: (i) on ETTh2 and ETTm2 at $n=10k$, a task-specific decodability benefit ($\Delta R^2 > 0$ in **10/10 CUDA seeds per dataset**; Ridge ΔR^2 rises from +0.12 to +0.67 $\pm$ 0.23; 5-hidden-layer MLP gives +0.70 $\pm$ 0.32, 10/10 positive); (ii) on ETTh1 at $n=10k$, forgetting resolves (10/10 negative, $-6.1\pm 3.6\%$) but ΔR^2 is 2/10 positive—a task-improvement-without-decodability-gain regime; and (iii) in the low- n regime, forgetting dominates. The ETTm2 result confirms the cost–benefit gap is independent of forgetting direction; the ETTh1 result narrows the $\Delta R^2 > 0$ invariant to ETTh2 and ETTm2. The benefit is *task-specific, not general*: $\Delta R^2 \leq 0$ on untrained heads (delta1 GBM: -0.056 ; forecast48: ≈ 0); the frozen-encoder control (condition D, 5 seeds, CKA=0.9993, $\Delta R^2 = -0.048\pm 0.008$, $14\times$ smaller than condition B) confirms the benefit requires encoder weight updates. Task forgetting is *protocol-conditional*: non-monotonic in n ; at $n=10k$, $-5.3\pm 6.2\%$ under CUDA 10-seed deterministic early-stopping (7/10 negative) vs. $+7.5\pm 8.2\%$ under final-epoch checkpointing. The $\Delta R^2 > 0$

majority is more robust than forgetting sign; forgetting is the downstream task-outcome consequence.

When does the cost–benefit gap hold? (a) Large ZS advantage and sufficient data: the CKA-probe gap is characterised in detail on (Moirai-Small, ETTh2) across $n \in \{500, \dots, 10k\}$; consistent low- n patterns hold on adjacent cells (ETTh1, ETTm2, Moirai-Base, Moirai-Large at $n=500$). Task forgetting resolves on (Moirai-Small, ETTh2) at $n=2k$ (-2.9%), and at $n=10k$ the CUDA 10-seed deterministic early-stopped protocol gives $-5.3 \pm 6.2\%$ (7/10 negative); final-epoch $k=5$ was $+7.5 \pm 8.2\%$, dominated by seed 101’s late-epoch overshoot ($+21.9\%$ collapsing to -20.7% at best-val-MSE epoch 4). Resolution does *not* transfer uniformly across ETT datasets: at $n=2,000$, ETTh1 still forgets ($+9.2 \pm 5.0\%$) while ETTm2 improves ($-20.1 \pm 5.1\%$). ETTh1 at $n=10k$ (CUDA $k=10$) gives a third pattern: task resolves (10/10 negative, $-6.1 \pm 3.6\%$) but ΔR^2 is mostly negative (2/10 positive), narrowing the per-seed $\Delta R^2 > 0$ invariant to ETTh2 and ETTm2. Full n -sweeps on Moirai-Base and Moirai-Large are future work. (b) Small or sub-random ZS (IoT, Weather): drift is benign or the gate is unmeasurable — IoT is a negative control, not a second dissociation instance. (c) Transition regime ($n \approx 5k$ on ETTh2-Small): 3/7 seeds overshoot in epoch 1 (§5.6), producing bimodal forgetting.

Limitations (App. A): dissociation on Moirai only — nine-of-nine non-Moirai backbone $\times$ ETT-dataset cells gate-fail (Chronos-T5-Small/Base/Large; TimesFM-2.5-200M; MOMENT-1-base); the Chronos size sweep confirms the ETT-coverage gap is size-invariant; Traffic/ILI and a gate-passing second backbone open. On the 96-step-ahead head, absolute $R^2(\text{FT})$ remains negative across all probes at every sampled n , so ΔR^2 there is a relative-improvement signal; the next-step-delta head with a GBM probe gives $R^2(\text{PT}) > 0$ but reveals the ΔR^2 gain is objective-specific (§5.5). A gate-passing non-Moirai backbone on Traffic or Exchange rate datasets (where non-ETT backbones are more likely to pass the value gate) is the highest-priority future-work direction for broadening the single-backbone scope.

A Limitations and Future Work

(1) Single-backbone scope of the dissociation claim. The drift-utility dissociation is established on the *Moirai* family only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTh2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.6%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTh2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213	−148%

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (−42.6% → −34.3% → −35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The central sample-size sweep (Table 3) covers ETTh2 at $n \in \{500, 1k, 2k, 5k, 10k\}$. At $n=10k$ we report the CUDA 10-seed deterministic early-stopped result: $k=10$ seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); `-deterministic` flag (CUBLAS_WORKSPACE_CONFIG=:4096:8, cudnn.deterministic=True, seeded DataLoader, PYTHONHASHSEED). Forg. $-5.3 \pm 6.2\%$, 7/10 negative; Ridge $\Delta R^2 = +0.67 \pm 0.23$, 10/10 positive; mean best epoch 4.2 ± 1.4 . The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the $n=5k$ overshoot pattern (§5.6). For comparison, MPS early-stopped $k=10$ gave $-5.7 \pm 10.4\%$ (8/10 neg, 9/10 $\Delta R^2 > 0$); a second early-stopped run ($k=5$) gave $-11.7 \pm 5.3\%$ (all 5 negative); final-epoch $k=5$ gave $+7.5 \pm 8.2\%$. Full n -sweeps on Moirai-Base and Moirai-Large are future work.

(3) Probe-head capacity. Ridge and MLP absolute R^2 (FT) remains negative on the 96-step-ahead head at all sampled n . The delta1/GBM probe gives R^2 (PT)=+0.059 but $\Delta R^2 = -0.056$ (§5.5). ΔR^2 on the 96-step head is a *relative* signal; Appendix H confirms its direction is not a probe-design artefact.

(4) Sample-sweep on other sizes. Moirai-Base/Large are evaluated at $n=500$ only; full n -sweeps are future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

B Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction — for ETTh2 at $h=192$, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper’s evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast’s central tendency.

C ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai’s `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai’s internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate $h=96$ and $h=192$ where Moirai’s pre-training advantage is 28% and 45% respectively. Horizons $h=336$ and $h=720$ exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

D LoRA Rank and Hyperparameter Sensitivity

Table 5 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve $\text{CKA} \geq 0.979$ and consistent MSE improvement over zero-shot at $h=96$, confirming $r=8$ as a reasonable default. On **Moirai-Large** (seed 42, $h=96$, $n=500$), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table 6 extends the sweep to α , target-module scope, and LR (all at $r=8$, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to 1×10^{-5} does (forg.= $-8.5 \pm 0.7\%$, $\text{CKA}=0.989 \pm 0.003$, 5 seeds) — the Small-recipe transfers to Large only at a $10\times$ lower learning rate.

E CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Table 5: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and $\text{CKA} \geq 0.979$. Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small’s negative forgetting. Mean $\pm$ std across 3 seeds (5 for $r=8$ Small); Moirai-Large single-seed ($s=42$) for $r \in \{16, 32, 64\}$, 3 seeds for $r=8$.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116 $\pm$.001	−9.5 $\pm$ 0.9	.992 $\pm$.001	3
Small	4	192	.146 $\pm$.003	−8.8 $\pm$ 1.9	.995 $\pm$.002	2
Small	8	96	.116 $\pm$.001	−9.1 $\pm$ 1.1	.979 $\pm$.006	5
Small	8	192	.157 $\pm$.002	−1.8 $\pm$ 3.1	.986 $\pm$.005	5
Small	16	96	.117 $\pm$.001	−10.2 $\pm$ 1.2	.987 $\pm$.002	3
Small	16	192	.155 $\pm$.002	−1.7 $\pm$ 1.2	.988 $\pm$.001	3
Small	32	96	.119 $\pm$.001	−8.3 $\pm$ 0.9	.986 $\pm$.001	3
Small	32	192	.159 $\pm$.003	+1.1 $\pm$ 1.4	.989 $\pm$.002	3
Large	8	96	.159 $\pm$.004	+25.7 $\pm$ 3.2	.973 $\pm$.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

Table 6: LoRA-Large HP ablation (ETTh2, $r=8$, $n=500$, $h=96$). Baseline (default LR=1e-4, $\alpha=16$, modules = q,v,out) forg. = +22.5%, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

Axis	Setting	Val MSE	Forg.%	CKA	Seeds
LR	1e-4 (default)	.157	+22.5	.955	1
	5e-5	.146	+15.3	.982	1
	1e-5	.115$\pm$.001	−8.5$\pm$0.7	.989$\pm$.003	5
α	8	.155	+21.9	.975	1
	16 (default)	.157	+22.5	.955	1
	32	.158	+24.3	.984	1
Modules	q,v,out (default)	.157	+22.5	.955	1
	q,k,v	.153	+20.4	.987	1
	q,k,v,out	.151	+19.0	.988	1

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 7.

The random-init CKA is $\leq 10^{-4}$ across 5 seeds (reported at 4 decimals as 0.0000 ± 0.0001), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

F Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head ($\alpha=1.0$) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same

Table 7: CKA calibration. Random-init CKA (0.000 ± 0.000 , 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.0000 ± 0.0001
Pre-trained vs. ETT fine-tuned (B, $h=96$)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 ± 0.026
Pre-trained vs. Frozen encoder (D)	1.000

format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. The probe training set uses $n_{\text{probe}}=300$ held-out examples; the remainder forms the test split used for MSE evaluation. ΔR^2 denotes fine-tuned minus pre-trained probe R^2 : positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder’s representational basis.

Key result. The sample-size sweep (Table 3) shows ΔR^2 monotonically increasing with training data (Ridge: $+0.12$ at 500 samples $\rightarrow +0.67 \pm 0.23$ at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS $k=5$ early-stopped gave $+0.56 \pm 0.14$ and final-epoch $k=5$ gave $+0.78 \pm 0.14$), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low n , drift destroys task-relevant features faster than it builds replacements; at higher n , the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that $R^2(\text{FT})$ remains *absolutely negative* at every sampled $n \in \{500, 1k, 2k, 5k, 10k\}$ (ZS: $R^2 = -6.90$; CUDA $k=10$ early-stopped: -6.79 at $n=500 \rightarrow -6.24 \pm 0.24$ at $n=10k$). $\Delta R^2 > 0$ is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains $\text{forg.} < 0$ at $n \geq 2k$. **Cross-dataset ΔR^2 comparability.** Raw ΔR^2 is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 ($n=10k$, CUDA $k=10$), the ZS Ridge floor is $R^2(\text{ZS}) = -24.52 - 3.5 \times$ deeper than ETTTh2’s -6.90 — which proportionally inflates the raw ΔR^2 ($+5.43$ on ETTm2 vs. $+0.67$ on ETTTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive ΔR^2 .

Horizon-sweep probe: head-specificity of the ΔR^2 gain. Table 8 reports Ridge ΔR^2 across three probe heads on four early-stopped encoders at $n=10k$ (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head: $\Delta R^2 = +0.116 \pm 0.153$ ($k=4$). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for), $\Delta R^2 \approx 0.000 \pm 0.163$ — the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives $R^2(\text{PT}) = +0.059 > 0$, establishing a positive-floor reference, with $\Delta R^2 = -0.056 \pm 0.030$ (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§5.5).

Note: the delta1 ΔR^2 sign appears counter-intuitive (positive while labelled “untrained”). The large positive delta1 ΔR^2 reflects that the ZS delta1 Ridge baseline is very negative ($R^2(\text{ZS}) \approx -2.3$) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§5.5): GBM gives $R^2(\text{PT}) = +0.059 > 0$ and $\Delta R^2 = -0.056$, yielding $R^2(\text{FT}) = +0.003$ — the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note: $R^2(\text{PT})$ varies slightly across GBM probe depths (depth 4: $+0.062$, depth 6: $+0.059$, depth 8: $+0.054$) because the probe is re-fit independently at each depth; this reflects probe capacity, not

Table 8: Ridge ΔR^2 by probe head on early-stopped encoders ($n=10k$, seeds {303, 777, 888, 999}). “Forecast96” is the trained head; “Forecast48” and “Delta1” are untrained. The gain is head-specific to the trained objective.

Head	Trained?	Ridge ΔR^2 per seed			
		s303	s777	s888	s999
Forecast96 (96-step ahead)	✓	+0.092	−0.087	+0.266	+0.192
Forecast48 (sub-horizon)	×	+0.061	−0.111	−0.156	+0.199
Delta1 (next-step delta)	×	+1.248	+0.757	+1.191	+0.980
Mean $\pm$ std (forecast96)			+0.116 $\pm$ 0.153		
Mean $\pm$ std (forecast48)			−0.001 $\pm$ 0.163		
Mean $\pm$ std (delta1)			+1.044 $\pm$ 0.223		

encoder structure.

ETTh2 delta1 GBM reprobe. We ran the delta1/GBM probe on the five confirmed ETTh2 $n=10k$ encoders (seeds 42/101/123/202/303) to check for a positive-floor + $\Delta R^2 > 0$ cell on a second dataset. Result: $R^2(\text{ZS}) = -0.030$ (negative floor) and per-seed $\Delta R^2 = \{+0.025, -0.060, +0.026, +0.030, +0.016\}$ (4/5 positive, mean $+0.007 \pm 0.031$). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM, $R^2(\text{ZS}) = +0.059$) gives $\Delta R^2 = -0.056$ — no positive-floor + $\Delta R^2 > 0$ cell was found. The $\Delta R^2 > 0$ signal on the trained 96-step head operates in a deeply-negative- R^2 regime; see Appendix H for probe noise-floor analysis.

Per-seed breakdown at $n=10k$ (CUDA 10-seed). Ridge ΔR^2 (all positive; mean $+0.668 \pm 0.226$): 42: $+0.205$; 101: $+0.736$; 123: $+0.937$; 202: $+0.604$; 303: $+0.793$; 456: $+0.823$; 777: $+0.344$; 789: $+0.873$; 888: $+0.561$; 999: $+0.804$. Absolute $R^2(\text{FT})$: all in $[-6.70, -5.97]$ ($R^2_{\text{ZS}} = -6.904$). MLP $k=5$ (all positive; mean $+0.701 \pm 0.324$): 42: $+0.979$; 101: $+0.762$; 123: $+0.948$; 202: $+0.771$; 303: $+1.007$; 456: $+0.266$; 777: $+0.076$; 789: $+0.847$; 888: $+0.449$; 999: $+0.901$.

ETTh2 $n=10k$ absolute R^2 per seed. The ETTh2 ZS Ridge floor ($R^2(\text{ZS}) = -24.52$, same pre-trained encoder across all seeds) is $3.5\times$ deeper than ETTh2’s -6.90 , explaining the raw ΔR^2 magnitude difference. Per-seed absolute values for the five available CUDA seeds:

Seed	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2	Forg. %
42	−24.52	−14.13	+10.39	−26.8
101	−24.52	−15.61	+8.91	−27.4
123	−24.52	−19.87	+4.64	−31.1
202	−24.52	−20.83	+3.69	−28.8
303	−24.52	−20.41	+4.11	−29.5
Mean (5 seeds)	−24.52	−18.17 $\pm$ 3.08	+6.35 $\pm$ 3.08	−28.7 $\pm$ 1.6

All five seeds show $R^2(\text{FT}) > R^2(\text{ZS})$ (i.e., fine-tuned representations are *less* linearly misaligned with the ETTh2 forecasting target than pre-trained ones), and all show $\Delta R^2 > 0$. The cross-dataset comparable statistic is the sign test: both ETTh2 (**10/10**) and ETTh2 (**5/5** seeds: 42, 101, 123, 202, 303) give unanimous $\Delta R^2 > 0$.

MLP depth sweep on early-stopped encoders. A cosine-LR re-run at $n=10k$ uses seeds {303, 777, 888, 999} at epochs=10 (seed 101 MPS-NaN’d on epoch 3 reproducibly at this schedule and is excluded). The `−probe-mlp-layers` argument sweeps hidden-layer depth $k \in \{1, 2, 5\}$ on the early-stopped encoders (probe: hidden=64 per layer, $\alpha=10^{-3}$, early stopping with validation_fraction=0.1, max_iter=500):

Table 9: Per-seed results at $n=5,000$ (ETTh2, $h=96$, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	R^2 (FT)
42	.132	+4.8	.531	+.67	-6.24
101	.120	-4.9	.571	+.75	-6.16
123	.124	-1.4	.544	+.81	-6.10
202	.128	+1.4	.461	+.15	-6.76
303	.150	+19.0	.525	+.52	-6.39
456	.153	+21.4	.711	+.20	-6.70
789	.134	+5.9	.480	+.40	-6.51
Mean $\pm$ std	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	+.50 $\pm$.26	-6.41 $\pm$.26

Seed	Ridge ΔR^2	MLP $k=1$	MLP $k=2$	MLP $k=5$
303	+0.092	+1.306	+1.418	+0.540
777	-0.087	+1.136	+1.554	+0.759
888	+0.266	+2.004	+2.361	+1.034
999	+0.192	+1.816	+1.949	+0.908
Mean $\pm$ std	+0.116 $\pm$ 0.132	+1.566 $\pm$ 0.356	+1.821 $\pm$ 0.368	+0.810 $\pm$ 0.184

The cosine-LR Ridge ΔR^2 is smaller than the epochs=20 run (+0.56 $\pm$ 0.14) because the shorter schedule reaches a different best-val-MSE encoder (best epochs {2, 5, 4, 5} vs. {2, 4, 8, 4, 3}), and Ridge is most sensitive to the specific linear basis. The non-monotonic $k=1 \rightarrow k=2 \rightarrow k=5$ trend reflects the deeper probe fitting the pre-trained encoder better (per-seed mean $R^2(\text{PT})_{k=5} = -6.93$ vs. $R^2(\text{PT})_{k=1} = -8.48$), shrinking the Δ from above, not degraded fine-tuned encoding. All four seeds are individually positive for $k=5$ MLP (+0.54, +0.76, +1.03, +0.91), preserving the cost-benefit gap. This run’s CKA is 0.592 $\pm$ 0.142 (mean best epoch 4.0 $\pm$ 1.22, higher than the earlier run’s 4.2 $\pm$ 2.3 because of the earlier LR annealing); forgetting is -7.13 $\pm$ 1.34%, tighter than the epochs=20 run’s -11.7 $\pm$ 5.3% and fully consistent with the $\text{forg.} < 0$ finding.

Crossover regime: per-seed breakdown at $n=5,000$. Table 9 reports all 7 seeds individually at $n=5,000$. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forgetting (-4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8% to +21.4%, mean +15.1%). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55) varies less than forgetting. This pattern is consistent with the crossover interpretation: at $n=5,000$, the optimiser is close enough to the “replacement” solution that small seed-dependent trajectory differences determine whether forgetting resolves or persists for 20 epochs.

F.1 $n=5,000$ per-epoch trajectory analysis

To support the mechanistic claim in §6, we plot the per-epoch training trajectories of all seven $n=5,000$ seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val MSE reaches its minimum separates the modes (0.7 $\pm$ 0.5 vs. 6.5 $\pm$ 2.1); mid-training weight drift (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

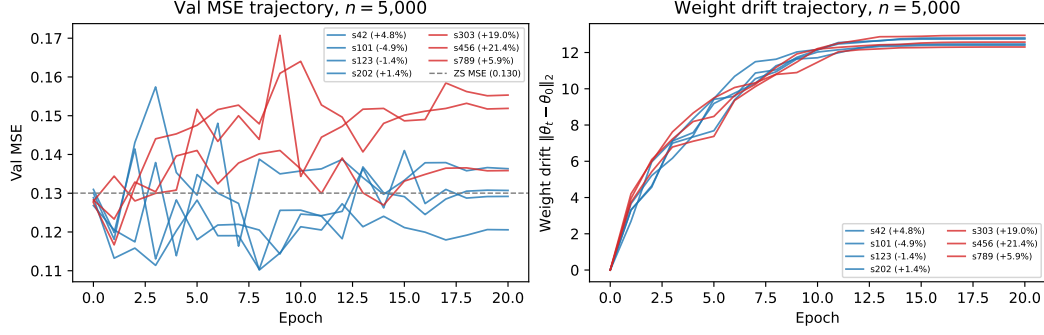


Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven $n=5,000$ seeds; both panels carry per-seed labels (sXXX (forg. %)) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq 5\%$); red: high-forgetting mode (forg. $> 5\%$). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

G IoT Negative-Control Details

The IoT negative-control evaluation uses CICIoT2023 [21] with 12 network-flow features and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the 95% CI of Moirai’s forecast over a 32-timestep scored horizon. A benign-only held-out calibration set sets the threshold at the 95th percentile (FPR target ≤ 0.05). All Moirai IoT conditions are trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

CNN-NLL from-scratch control. To directly address the foundation-model justification at IoT’s 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-NLL ($\text{AUC}=0.598\pm0.041$) matches fine-tuned Moirai B/D ($\text{AUC}=0.556\pm0.021$) and is close to the best Moirai condition C ($\text{AUC}=0.629\pm0.022$). A CNN-MSE variant ($\text{AUC}=0.580\pm0.023$) rules out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent with Moirai’s pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale. This is the expected outcome when pre-trained features carry no task-relevant signal — a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

H Probe Noise-Floor Analysis

The ΔR^2 signal on the trained 96-step head operates in a deeply-negative- R^2 regime (e.g., ZS -6.90 , FT -6.24 at $\alpha=1$). We test whether the $\Delta R^2 > 0$ direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2, $n=10k$) across: (1) $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, (2) probe seeds $\{0, 1, 2\}$, (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

Results (Table 10).

Finding: $\Delta R^2 > 0$ holds for all 10 probe-design settings tested; the direction is not an artefact of α , pooling, or probe seed. Magnitude decreases with stronger regularisation (larger α shrinks both R^2 values toward zero), but the sign is invariant. The absolute R^2 values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust.

Table 10: ΔR^2 under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All $\Delta R^2 > 0$.

Setting	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2
$\alpha=0.01$	-13.51	-11.54	+1.96
$\alpha=0.1$	-8.78	-7.94	+0.84
$\alpha=1.0$ (default)	-6.90	-6.70	+0.20
$\alpha=10$	-5.81	-5.72	+0.08
$\alpha=100$	-5.32	-5.28	+0.04
Probe seeds 0/1/2 ($\alpha=1$, mean pool)	same as $\alpha=1.0$		+0.20
Last-token pool ($\alpha=1$)	-9.70	-8.49	+1.20

I Layer-Wise Unfreeze Localization

Moirai-Small has 12 transformer layers. We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2 $n=10k$, $h=96$, 3 seeds (42/101/123) at $N=3$ alongside the existing $N=0$ (frozen encoder, 5-seed condition D) and $N=12$ (full unfreeze, condition B).

$N=1$ instability. At $N=1$, training diverged (NaN activations at epoch 11, batch 148) with default $\text{LR} = 10^{-4}$, despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at ≈ 0.72 ; we do not extract probe results from a crashed run. This instability is qualitatively consistent with the LoRA-Large behaviour documented in §5.4: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

$N=3$ results. Table 11 shows results for $N=0$, $N=3$ (3 seeds), and $N=12$ (3 matching seeds).

Table 11: Layer-wise unfreeze profile (ETTh2 $n=10k$, $h=96$; mean $\pm$ std over seeds). N = number of top transformer layers unfrozen. $N=0$: 5 seeds; $N=3/N=12$: seeds 42/101/123.

N	Layers	CKA	Ridge ΔR^2	Forgetting%
0 (frozen)	none	0.999 ± 0.000	-0.048 ± 0.008	$-6.6 \pm 1.5\%$
1 (top-1)	1/12	≈ 0.72 (ep. 11)	<i>diverged (NaN)</i>	
3 (top-3)	3/12	0.698 ± 0.066	$+0.479 \pm 0.115$ (3/3)	$+1.3 \pm 2.1\%$
12 (full, B)	12/12	0.484 ± 0.061	$+0.626 \pm 0.378$ (3/3)	$-7.7 \pm 8.4\%$

Finding. $\Delta R^2 > 0$ appears already at $N=3$ (3/3 seeds), with less encoder drift (CKA ≈ 0.70) than full unfreeze (CKA ≈ 0.48). The top-3 layers are sufficient to produce the relative decodability gain; the remaining 9 frozen layers contribute additional drift without consistently increasing ΔR^2 (mean $+0.63$ vs. $+0.48$, but with larger variance 0.378 vs. 0.115). Both $N=3$ and $N=12$ operate in the same deeply-negative- R^2 regime ($R^2(\text{ZS}) \approx -6.9$); the same probe-design robustness established in Appendix H applies.

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